

TIANTONG WANG

📍 Nanyang Technological University

✉ tiantong001@e.ntu.edu.sg

☎ (+86)18912133505 / (+65)85354400

EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHKSZ)

Shenzhen, China

Bachelor of Science in Mathematics and Applied Mathematics

Sept. 2019 - June 2023

Stream: Applied Mathematics

GPA: 3.794/4.000 Top 6% (School), Top 9% (Major)

Honors: Academic Performance Scholarship in Academic Year 2019-2020 & 2020-2021

(Ranked *1st among 551 students* in School of Science and Engineering in academic year 2019-2020)

Curriculum: *Mathematical Introduction to Deep Learning, Optimization I & II, Optimization Models and Methods in Deep Learning, Mathematical Modeling, Ordinary Differential Equations, Partial Differential Equations, Numerical Analysis, Calculus I & II, Linear Algebra, Advanced Linear Algebra, Real Analysis, Abstract Algebra, Graph Theory, Applied Category Theory*

Cambridge University

Cambridge, UK

Online Summer Research Program (Physics), Pembroke College

July 2022

Supervisor: Joao Rodrigues

Research Topic: Analysis of the Motion of a Two-Dimensional Particle-Spring System with Friction Dependent on the Velocity

Honor: Full scholarship co-funded by Cambridge and CUHKSZ

Nanyang Technological University

Singapore

PhD in College of Computing and Data Science

Aug. 2023 - present

Joint NTU-WeBank Research Centre on Fintech (2023-2024)

Alibaba-NTU Global e-Sustainability CorpLab (ANGEL) (2025-present)

Supervisors: Lim Wei Yang Bryan, Chunyan Miao

Research Topics: Collaborative AI system, model merging.

PUBLICATIONS

- Artificial Intelligence without Restriction Surpassing Human Intelligence with Probability One: Theoretical Insight into Secrets of the Brain with AI Twins of the Brain. *Neurocomputing*, Feb 2025
- AegisGuard: RL-Guided Adapter Tuning for TEE-Based Efficient & Secure On-Device Inference. *Neurips* 2025.
- M-Loss: Quantifying Model Merging Compatibility with Limited Unlabeled Data. *AAAI* 2026.

RESEARCH EXPERIENCE

Weekly Seminars on Advanced Data Science Topics

Sept. 2021 - Dec. 2021

Supervisor: Prof. Zhi-Quan Tom Luo, Academician of Chinese Academy of Engineering

- Discussed with other five distinctive students and Prof. Luo weekly on data science topics, such as Markov chains, graph theory, and machine learning
- Gave presentations and hosted the discussion on the topic *Machine Learning*
- Provided a **new proof** of a theorem regarding VC-Dimension in machine learning
- **Final assessment with grade A**

Convergence Analysis of Consensus Based Decentralized Federated Learning Algorithms Jan. 2022 - Aug. 2022
Network Communications and Economics Lab, the Chinese University of Hong Kong, Shenzhen

Supervisors: Prof. Jianwei Huang, Dr. Bo Liu

- Mathematically analyzed the convergence rate and properties of a consensus based decentralized learning algorithm

- Conjectured and proved a key theorem regarding the convergence property of the algorithm in Federated Learning
- **Final oral defense and report with grade A**

Analysis of the Motion of a Two-Dimensional Particle-Spring System with Friction Dependent on the Velocity

July 2022

Online Summer Research Programme, Pembroke College, Cambridge

Supervisor: Dr. Joao Rodrigues

- Mathematically derived some formulae for the motion of a two-dimensional particle-spring system with a centripetal force proportional to distance and different viscous frictions dependent on velocity
- Conducted simulations of motions regarding different viscous frictions via MATLAB
- **Honor: Full scholarship** co-funded by Cambridge and CUHKSZ

ACADEMIC ACTIVITIES

Undergraduate Student Teaching Fellow Program

Sept. 2020 - June 2023

The Chinese University of Hong Kong, Shenzhen

Undergraduate Student Teaching Fellow

- Served as undergraduate teaching fellow for the following courses:

PHY1001: Mechanics

STA2001: Probability and Statistics

MAT3004: Abstract Algebra

MAT3040: Advanced Linear Algebra

- Designed and gave tutorial classes weekly.
- Held office hours answering students' questions.

Online Summer Camp: Federated Learning, School of Science and Engineering

June 2020 - Aug. 2020

Supervisor: Prof. Jianwei Huang

- Learned some popular algorithms and paradigms of federated learning
- Conducted a project simulating the process of federated learning
- Trained a model with CIFAR10 data set. Got accuracy 86% in 3 training iterations and 98% in 100 training iterations

EXTRACURRICULAR ACTIVITIES

Fantastic Classes after School, Shimmer Public Welfare

Aug. 2019 - May 2020

Volunteer Manager

- Collaborated with three different primary schools and offered after-school activities biweekly for primary school students, such as interesting physics experiments
- Interviewed voluntary teachers and arranged the activity schedule and topics
- **Honors:** People's 100 Favorite Programs in Shenzhen, Outstanding Programs in Guangdong-Hong Kong-Macao Greater Bay Area Benefiting the Children

Student Representative, the Chinese University of Hong Kong, Shenzhen

Sept. 2019 - July 2021

Student Representative for IT Security (2019-2020)

Student Representative for Shall College student welfare (2020-2021)

- Work with university IT department on formalization of IT privacy policy
- Work with Shall College (Hall) office for the replacement of old hall facilities.

Physics Association, the Chinese University of Hong Kong, Shenzhen

Sept. 2020 - July 2021

Vice-President, **Founder**

- Held seminars for students to discuss advanced topics on physics
- Invited professors to give lectures on advanced topics
- Invited distinguished students to share experience on physics study

Table Tennis Team, Shaw College

Sept. 2022 - June 2023

Team Manager

- Arranged regular training and other activities, such as table-tennis opening day, inter-college table-tennis matches
- Financial management